

Guy Kaplan

PhD Student, Computer Science · The Hebrew University of Jerusalem

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Education

The Hebrew University of Jerusalem

PhD Student, Computer Science

Jerusalem, Israel
Aug. 2025 – Present

Advisor: Prof. Roy Schwartz, School of Computer Science & Engineering.
Research: inner lexicon & token-level representations of LLMs; mechanistic interpretability.

The Hebrew University of Jerusalem

M.Sc., Computer Science (GPA: 95/100)

Jerusalem, Israel
Oct. 2022 – Aug. 2025

Advisor: Prof. Roy Schwartz, School of Computer Science & Engineering.
Thesis: *The Inner Lexicon of Large Language Models* — published at ICLR 2025.

The Open University of Israel

B.Sc., Computer Science (GPA: 90/100, Dean's Honours 2021)

Israel
Oct. 2019 – Jun. 2022

Tel Aviv University

B.A., Economics

Tel Aviv, Israel
Oct. 2019 – Jun. 2022

Concurrent with B.Sc. studies at the Open University of Israel.

University of Haifa

B.A., General Studies, Magna Cum Laude (GPA: 93/100)

Haifa, Israel
2014

Research & Professional Experience

University of Southern California (USC)

Visiting Researcher, DILL Lab (Prof. Swabha Swayamdipta)

Los Angeles, CA
Sep. 2025 – Present

- Full-year official research visit; active collaborations on reward hacking in LLMs and mechanistic interpretability.

Microsoft

Data Scientist

Israel
Mar. 2024 – Jan. 2025

- Built large-scale NLP and anomaly-detection models for risky-user detection.

Bright Forensic Innovations

Chief Scientist Officer

Tel Aviv, Israel
Oct. 2023 – Present

- Developed AI method for cadaver identification from dental records; operationally deployed by the Israeli Police for October 7 victim identification; raised top-10 accuracy from 10% to 89%.
- Lectured the Israeli Police forensic unit on the AI methodology.

Microsoft

Software Engineer

Israel
Mar. 2022 – Mar. 2024

Built EventHub & Cosmos DB features for the Azure Defender UEBA product; drove CI/CD rollout.

Israeli Navy — Air Patrol Unit

Naval Officer (Major)

Oct. 2011 – Sep. 2019

Commanded 50 personnel; earned Palmachim AFB Commander's Award.

Publications

- [1] *From Tokens to Words: On the Inner Lexicon of LLMs*
Guy Kaplan, Matanel Oren, Yuval Reif, Roy Schwartz

International Conference on Learning Representations (ICLR), 2025 (presented in Singapore).
arxiv.org/abs/2410.05864

- [2] *Follow the Flow: On Information Flow Across Textual Tokens in Text-to-Image Models*
Guy Kaplan*, Michael Toker*, Yuval Reif, Yonatan Belinkov, Roy Schwartz (*equal contribution)
Annual Meeting of the Association for Computational Linguistics (ACL), 2026
arxiv.org/abs/2504.01137
- [3] *Vocab Diet: Reshaping the Vocabularies of LLMs with Vector Arithmetic*
Yuval Reif, **Guy Kaplan**, Roy Schwartz
Findings of the Association for Computational Linguistics (ACL Findings), 2026
arxiv.org/abs/2510.17001
- [4] *Why Fine-Tuning Encourages Hallucinations and How to Fix It*
Guy Kaplan, Zorik Gekhman, Zhen Zhu, Yuval Reif, Lotem Rozner, Swabha Swayamdipta, Derek Hoiem, Roy Schwartz
Conference on Language Modeling (CoLM), 2026
arxiv.org/abs/2604.15574
- [5] *More Than Words: Compositional Tokenization for Efficient Language Models*
Yuval Reif, **Guy Kaplan**, Roy Schwartz
Conference on Language Modeling (CoLM), 2026
openreview.net/pdf?id=Yw16kddexd

Teaching & Mentoring

- Mentored Omer Ben Shahar (undergraduate, 2025) on training-data frequencies and the inner lexicon of LLMs; manuscript in preparation for ARR submission.
- Mentored Lotem Rozner (M.Sc.) on LLM personalization, extending the inner-lexicon research program; her work resulted in co-authorship on *Why Fine-Tuning Encourages Hallucinations and How to Fix It*.

Invited Talks

"The Inner Lexicon of LLMs" University of Southern California (USC)	Jan. 2026
"The Inner Lexicon of LLMs" Tel Aviv University (TAU)	Dec. 2025
"The Inner Lexicon of LLMs" Microsoft Israel Data Science Group	Jan. 2025

Awards & Honors

PBC (VATAT) Fellowship for Outstanding Doctoral Students in AI & Data Science Council for Higher Education of Israel, under the National Initiative for Artificial Intelligence and Data Science. Israel's national doctoral fellowship in AI and data science: up to 20 awards countrywide per cycle, with each university permitted at most six nominees; three years of full doctoral funding.	2026
Excellence Graduate Student Scholarship, KLA	2025
Dean's Honours in B.Sc. Computer Science, Open University of Israel	2021
Commander's Award, Palmachim AFB, Israeli Navy	2019
Magna Cum Laude, B.A. General Studies, University of Haifa	2014

Academic Service

Reviewer: ACL Rolling Review (ARR), Conference on Language Modeling (CoLM), ICML 2024 Workshop on Actionable Interpretability

Research Collaborations: Technion, UIUC, Google Research, ETH Zürich (official research visit), University of Southern California

Volunteering

Israeli Scouts — Bedouin Youth at Risk & Jewish–Arab Coexistence

Year of Service Volunteer

Negev, Israel

2010 – 2011

Full-time educational & social engagement with at-risk Bedouin youth; promoted Jewish–Arab coexistence.

Skills

ML & NLP: Transformers, mechanistic interpretability, LLM fine-tuning (LoRA/QLoRA), contrastive learning

Programming: Python (PyTorch, TensorFlow), Java, Bash, L^AT_EX

Languages: Hebrew (native), English (fluent)